Lesson 26: The Single-Subject Versus Group Debate

English Stage 4 (Psychology) · Unit 3: Research Designs

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 10.4: The Single-Subject Versus Group "Debate"; 10.5: Single-Subject Research (Summary).

In this lesson

You will learn why single-subject and group researchers disagree about data analysis and external validity, why the approaches are best seen as complementary, and the principle of converging evidence. In English, you will practise the moves of academic argument: stating a concern, conceding it and answering it.

Learning goals

After this lesson you should be able to:

- Explain the disagreements about data analysis and external validity, and each side's response.
- Decide whether a research question suits single-subject or group research.
- Explain the principle of converging evidence.
- Present both sides of a methodological debate with concession and counter-argument.

Key language

Term or phrase	Meaning	Example
group research	Studying many participants and examining behaviour in terms of means, standard deviations and other group-level statistics.	A between-subjects experiment
visual inspection	Graphing the data and judging whether, and how much, the independent variable affected the dependent variable.	Researchers may disagree about one graph.
steady state strategy	Letting behaviour become fairly consistent before changing conditions, so an effect is easier to detect.	It reduces "noise" in the data.
external validity	The extent to which results can be generalized to people and situations beyond those actually studied.	Two children with intellectual disabilities
replication	Conducting a study again, exactly or with modifications, to see if the same results are observed.	A new small sample of slightly different participants

Term or phrase	Meaning	Example
bimodal distribution	A distribution with two peaks, visible in a histogram (a graph of how often each score occurs).	Some participants helped, others harmed
converging evidence	Judging a body of research by its pattern of flaws: agreement among studies with *different* flaws increases confidence.	Correlational and experimental studies agree
share this concern	Concedes that the other side's worry is valid	"Again, single-subject researchers share this concern."
turning the tables	Introduces criticism of the critics	"Turning the tables, some advocates of single-subject research worry..."
it is probably best to conceptualize X as Y	A hedged conclusion	"...as complementary methods"

Rules and explanations

The ideas

Common ground. Both approaches are quantitative and try to establish causal relationships by manipulating an independent variable, measuring a dependent variable and controlling extraneous variables. Their differences lead to disagreements.

Who worries about what	The concern	The reply
Group researchers: visual inspection	It may miss weak effects; it can be unreliable, with different researchers reaching different conclusions; its results cannot be clearly summarized or compared across studies.	Shared concerns, but the steady state strategy and the focus on strong, consistent effects minimize most of them. Unclear effects are strengthened or the noise reduced; if still unclear, they are not of interest. Statistics increasingly supplement it.
Single-subject researchers: group means	A strong positive effect on half the participants and an equally strong negative effect on the other half cancel out, so the treatment seems to have no effect.	Shared concern. Group researchers examine distributions of individual scores (a bimodal histogram), use within-subjects designs to see individual effects, and use factorial designs to compare groups (introverts vs. extraverts).
Group researchers: small samples	Will a treatment that reduced self-injury in two children work for other children?	Shared concern, but strong, consistent effects are likely to generalize; results are replicated with other small samples; conditioning principles, found mostly with single-subject research, generalize across many species and situations.

Who worries about what	The concern	The reply
Single-subject researchers: group results	A small average effect does not tell us whether the next *individual* will show a small, a large or even a negative effect. Group studies also generalize from one situation (cell phone use on a closed oval track).	The book concludes that generalizing depends on the similarity of the participants *and* situations studied to those of interest, not just on the number of participants.

Complementary methods.

- *Single-subject research* suits testing treatments on individuals when effects are strong, consistent and biologically or socially important; studying particular individuals; and clinicians who work with one client at a time.
- *Group research* suits testing treatments at the group level; detecting weak effects (which may lead to better treatments); studying interactions between treatments and participant characteristics (motivation to change); and studying variables that cannot be manipulated (number of siblings, extraversion, culture).
- *Research traditions* are "probably the most important" factor in the choice, but they can be integrated: single-subject research with rats and birds and group research with human infants show similar abilities to discriminate small numbers, an innate "number sense" (Dehaene, 2011).

Converging evidence. No design is perfect: true experiments typically have high internal validity but may have problems with external validity; correlational research often has the opposite profile. If consistent studies all share one flaw (for example, all are correlational), the agreement may come from the flaw. If their flaws differ, so that one study's weakness is another's strength, confidence increases. "In science, we strive for progress, not perfection."

Language focus

1. Concern, concession, counter-argument. The section repeats one pattern:

- Concern: "Some advocates of group research *worry that* visual inspection is inadequate..."
- Concession: "*In general*, single-subject researchers *share these concerns*."
- Counter-argument: "*However, they also argue that* their use of the steady state strategy ... minimizes most of them."
- Switch sides: "*Turning the tables*, some advocates of single-subject research worry..."

Inside one sentence: "*Although* they *do* focus on group statistics, they also emphasize the importance of examining distributions of individual scores." Emphatic *do* admits the point; the main clause answers it.

2. Hedged conclusions. "single-subject research and group research are *probably best* conceptualized as complementary approaches"; the number sense "*may even be* the foundation of humans' advanced mathematical abilities."

> **Note for Arabic speakers** - In Arabic, a concessive clause is often followed by a word meaning *but*, so learners write "Although X, but Y". English uses one connector only: "Although X, Y" or "X, but Y".

Practice

1. Reading questions

1. What are the three concerns about visual inspection?
2. How can group researchers detect a treatment that helps some participants and harms others?
3. Why can a small positive average effect not tell a clinician what will happen to the next client?
4. Name three kinds of question for which group research is better.
5. According to the principle of converging evidence, when do consistent findings deserve *less* confidence?

2. From the book (10.5 Exercises)

1. Practice: "Find and read a published article in psychology that reports new single-subject research. (An archive of articles published in the Journal of Applied Behavior Analysis can be found at http://www.ncbi.nlm.nih.gov/pmc/journals/309/) Write a short summary of the study." (The link now redirects; see Watch online.)
2. Practice: "Design a simple single-subject study (using either a reversal or multiple-baseline design) to answer the following questions. Be sure to specify the treatment, operationally define the dependent variable, decide when and where the observations will be made, and so on." (a) "Does positive attention from a parent increase a child's tooth-brushing behavior?" (b) "Does self-testing while studying improve a student's performance on weekly spelling tests?" (c) "Does regular exercise help relieve depression?"
3. Practice: "Create a graph that displays the hypothetical results for the study you designed in Exercise 1. Write a paragraph in which you describe what the results show. Be sure to comment on level, trend, and latency." (The book means the design exercise, item 2 here. Draw the graph on paper.)
4. Discussion: "Imagine you have conducted a single-subject study showing a positive effect of a treatment on the behavior of a man with social anxiety disorder. Your research has been criticized on the grounds that it cannot be generalized to others. How could you respond to this criticism?"
5. Discussion: "Imagine you have conducted a group study showing a positive effect of a treatment on the behavior of a group of people with social anxiety disorder, but your research has been criticized on the grounds that 'average' effects cannot be generalized to individuals. How could you respond to this criticism?"
6. Practice: "Redesign as a group study the study by Hall and his colleagues described at the beginning of this chapter, and list the strengths and weaknesses of your new study compared with the original study."

7. Practice: "The generation effect refers to the fact that people who generate information as they are learning it (e.g., by self-testing) recall it better later than do people who simply review information. Design a single-subject study on the generation effect applied to university students learning brain anatomy."

3. Language

1. Correct: "Although single-subject studies use few participants, but their results often generalize."
2. Join with *Although*: "Group researchers focus on group statistics. They also emphasize examining distributions of individual scores."

Speak with your tutor

- Role-play: your tutor plays a group researcher criticizing your single-subject study (Practice 2.4). Answer with concern, concession and counter-argument; then switch roles.
- Explain converging evidence with an example from an earlier lesson.
- PSYC 110 Lecture 5 (What Is It Like to Be a Baby: The Development of Thought) describes Karen Wynn's studies in which six-month-olds look longer when 2 – 1 = 2, suggesting rudimentary arithmetic. Connect this group research with infants to the book's "number sense" example.
- In the sleep part of PSYC 110 Lecture 17 (Self and Other, Part II; Some Mysteries: Sleep, Dreams, and Laughter), Professor Bloom notes big individual differences in how much sleep people need. Discuss what this means for a group study of sleep and memory.

Writing task

Your project: choosing between designs. Write 280-350 words comparing a group version of your study of sleep and memory (your experiment, or your quasi-experimental or factorial option) with your single-subject option from Lesson 25. On your own topic, compare the same two kinds of design.

Include: one strength and one weakness of each design for data analysis and for external validity; at least two concession and counter-argument moves; a hedged conclusion naming your main design and explaining how the other could provide converging evidence.

Self-editing checklist:

- I compared both designs on data analysis and external validity.
- Each concern is followed by a concession and a counter-argument.
- I used *Although X, Y* without *but*.
- My conclusion is hedged (*probably*, *may*).
- I explained how the second design could provide converging evidence.
- I used the book's terms (visual inspection, replication).

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- The *Journal of Applied Behavior Analysis* archive, 1968 to 2012 (the book's link redirects here): https://pmc.ncbi.nlm.nih.gov/journals/309/
- Hall, Lund and Jackson (1968), "Effects of Teacher Attention on Study Behavior" (free full text): https://pmc.ncbi.nlm.nih.gov/articles/PMC1310970/

Answer key

1. Reading questions

1. It may not detect weak effects; it can be unreliable; its results cannot be clearly and efficiently summarized or compared across studies (10.4).
2. By examining the distribution of individual scores: a histogram would show a bimodal distribution. Within-subjects designs also show individual effects.
3. Behind the average, some participants may show large positive effects and others negative ones, so one person's response cannot be predicted.
4. Questions about weak effects; interactions between treatments and participant characteristics; variables that cannot be manipulated.
5. When all the studies share the same flaw (e.g., all correlational), because the consistency may result from that flaw.

2. From the book (example answers)

1. (based on the book; add details from the article) Hall, Lund and Jackson (1968) observed six disruptive elementary school students every 10 seconds for 30 minutes a day. After a baseline, the teacher gave positive attention whenever a student worked. Study time rose sharply (Robbie: 25% to 71%), fell when the attention stopped and rose again when it returned (ABAB), showing that the attention caused the change.
2. (a) ABAB reversal, one child. Treatment: the parent praises the child right after brushing. Dependent variable: seconds spent brushing, timed by the parent in the bathroom each evening. Each phase lasts until brushing time is stable. (b) Multiple-baseline across three students, because learning may not return to baseline when self-testing stops. Treatment: self-testing during home study. Dependent variable: words spelled correctly out of 20 on the weekly class test. Self-testing starts after 3, 5 and 7 weeks of baseline. (c) Multiple-baseline across three adults, because removing an effective treatment could be unethical. Treatment: 30 minutes of brisk walking 5 days a week. Dependent variable: a weekly depression questionnaire score at the clinic. Exercise starts after 2, 4 and 6 weeks.
3. (for 2a) Brushing time was short and stable during the first baseline (about 30 seconds). It began to rise the evening after praise started (a short latency) and levelled off at about 2 minutes. When praise stopped, it fell gradually to about 45 seconds; it rose again quickly when praise returned. These changes in level and trend, with short latencies, suggest that praise caused the change.
4. I share this concern: one man may not represent everyone with the disorder. However, strong and consistent effects like this one are likely to generalize. I will also replicate the study with other small samples, for example women or people in other settings. Generalizing depends on the similarity of participants and situations, not only on sample size.
5. I share this concern: an average effect does not predict each person's response. However, the distribution of individual scores showed that most participants improved (it was not bimodal);

with pretest and posttest scores, I can report the percentage with strong, weak or negative effects; and factorial analyses can show for whom the treatment works.

6. Randomly assign disruptive students from many classrooms to positive teacher attention or to usual teaching (perhaps a waitlist control); measure time on schoolwork with the same 10-second method; compare group means statistically. Strengths: random assignment to a control group; weaker effects can be detected; a larger, more varied sample; interactions (e.g., with age) can be studied. Weaknesses: means may hide students who did not improve or got worse; far more costly; each student is measured less intensively.
7. An alternating treatments design with three students. Each day a student studies a new set of brain structures, alternating between self-testing (naming them from memory) and reviewing labelled diagrams. (It suits fast-acting treatments.) Dependent variable: the percentage of that day's structures named correctly on a quiz 24 hours later. A consistently higher level after self-testing days would support the generation effect.

3. Language

1. "Although single-subject studies use few participants, their results often generalize." (or "...use few participants, but their results often generalize.")
2. "Although group researchers focus on group statistics, they also emphasize examining distributions of individual scores."

My notes